GLOBE++ Lite-P+: Global-to-Local Policy Distillation with Predictive Risk Switching for Decentralized FANET Routing

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Abstract—Flying ad hoc networks (FANETs) require routing decisions that are both reliable under rapid topology changes and executable with only local information on resource-constrained unmanned aerial vehicles (UAVs). Centralized or communication-heavy multi-agent reinforcement learning methods can learn high-quality routing behavior, but their execution-time dependence on global state or inter-agent messages is difficult to deploy in sparse and mobile UAV swarms. This paper presents GLOBE++ Lite-P+, a decentralized routing framework that transfers routing knowledge from a privileged global teacher to a local student and augments the student with predictive risk switching. The teacher is trained offline with global topology information, while the deployed policy uses only one-hop candidate features, destination geometry, queue state, predicted link lifetime, onward stability, and lightweight safety gates. The proposed risk switch activates the predictive branch only when the nominal local decision is unsafe, thereby retaining the low-latency behavior of geographic routing in benign cases while improving recovery from routing holes and imminent link breaks. Full Phase 12 experiments over 14 scenarios and five random seeds show that Risk-Switch Lite-GLOBE-P, the fully evaluated predecessor of GLOBE++ Lite-P+, achieves the best overall packet delivery ratio and deadline delivery among GPSR, predictive geographic routing, Evo-QGeo, IQMR Q(λ), DRAMA, and internal Lite-GLOBE variants, while using fewer input bytes than predictive and message-passing RL baselines. The current Phase 13 GLOBE++ Lite-P+ implementation adds link-loss-aware switching, top-k onward stability, energy-aware tie breaking, and drop suppression; its full-scale validation is left as an explicit experimental TODO in this draft.

Index Terms—FANET, UAV routing, geographic routing, policy distillation, knowledge distillation, reinforcement learning, decentralized execution, routing holes

I. Introduction

UAV swarms are increasingly used as rapidly deployable communication platforms, sensing systems, and emergency networks. In such systems, packet routing is difficult because node positions change quickly, wireless links appear and disappear, and each UAV can normally observe only local neighborhood information. A routing policy must therefore satisfy two conflicting requirements: it should exploit enough topology and mobility structure to avoid unstable routes, while remaining lightweight enough for decentralized execution.

Classical geographic routing, represented by GPSR-style next-hop selection, is attractive because it is simple, local, and low overhead. However, purely greedy geographic forwarding can fail in routing holes, local dead ends, and high-mobility cases where the closest neighbor to the destination is about to lose connectivity. Reinforcement-learning and multi-agent reinforcement-learning approaches can incorporate richer information, but many of them require online message passing, centralized state, or expensive execution-time coordination. This creates a deployment gap: policies that know enough to route well are often too heavy to execute, while policies that are cheap enough to execute often lack global routing awareness.

This work studies that gap through a global-to-local design. Instead of claiming that global context itself is novel, we use global context only as a training-time privilege. A global teacher observes the full FANET graph during offline learning and dataset generation. A local student then learns a deployable next-hop policy from teacher and oracle-style routing targets. At execution time, the student no longer queries the teacher or a centralized controller. This distinction is central: the contribution is not online cooperation, but the transfer of routing structure into a local decision rule.

The final proposed algorithm in this draft is GLOBE++ Lite-P+. It builds on the fully evaluated Risk-Switch Lite-GLOBE-P model and adds four mechanisms motivated by Phase 12 failure analysis: link-loss-aware risk switching, top-k onward stability, energy-aware tie breaking, and drop suppression. These mechanisms are intentionally local. They do not require global routing commands, multi-hop control messages, or online global graph aggregation.

The main contributions are as follows:

- We formulate decentralized FANET routing as a partial-observation next-hop decision problem and separate training-time privileged information from execution-time local information.
- We design a global-to-local policy distillation pipeline in which a global graph teacher provides soft action distributions and routing targets to a local student.
- We propose predictive risk switching, a lightweight local safeguard that switches from the nominal distilled policy to a predictive branch only when link margin, link lifetime, onward stability, redundancy,

or link survival probability indicates elevated risk.
- We provide a full Phase 12 evaluation across 14 scenarios, five seeds, and six evaluated methods, and merge the results with external baselines GPSR, Predictive Geographic, Evo-QGeo, IQMR Q(λ), and DRAMA.
- We explicitly identify the remaining evidence gaps for a submission-quality GLOBE++ Lite-P+ paper, including Phase 13 full validation, AODV/OLSR-style classical baselines, statistical tests for P+ ablations, and full energy-model calibration.

This draft should be read as a paper-ready technical manuscript scaffold. Results reported for Phase 12 are supported by local experiment artifacts. Results for the Phase 13 P+ extension are not yet claimed as full-scale evidence.

II. Related Work

A. FANET and Geographic Routing

FANET routing protocols must respond to high mobility, intermittent connectivity, and limited communication range. Geographic routing methods such as GPSR [1] use node positions to forward packets toward the destination with minimal signaling. Their simplicity makes them strong baselines for UAV routing, but greedy progress can fail near void regions or when the geographically attractive next hop is unstable.

B. Predictive Geographic Routing

Predictive geographic protocols extend greedy routing by estimating link duration, neighbor mobility, or future link-state evolution. Evo-QGeo-style methods are particularly relevant because they consider future link-state evolution and routing-hole bypass [2]. This overlaps with any claim based only on “using predicted links”. For this reason, our novelty claim is not merely predictive routing, but a global-to-local teacher-student transfer combined with communication-free local execution and risk-triggered predictive switching.

C. Reinforcement Learning and MARL Routing

RL routing methods learn routing policies from rewards such as delivery success, delay, hop count, queue pressure, or link stability. Multi-agent RL methods can model distributed decisions, but execution-time cooperation can introduce signaling cost. DRAMA-like emergent-communication routing is therefore a close baseline because it explicitly uses learned inter-agent communication [3]. In contrast, the proposed method uses privileged global information offline and removes the online global teacher during deployment.

D. GNN-Based Routing and CTDE

Graph neural networks are natural for routing because topology can be represented as a graph. Centralized training with decentralized execution (CTDE) is also well established in multi-agent systems, including UAV-assisted networking studies such as GLo-MAPPO [4]. Consequently, the paper does not claim CTDE alone as the contribution. The routing-specific contribution is the distillation of global graph knowledge into local next-hop utilities and the use of local predictive safeguards during execution.

TABLE I
Conceptual positioning of routing methods.

Method family	Prediction	Online comm.	Local execution
GPSR	No	Low	Yes
Predictive geographic	Yes	Low/medium	Yes
Evo-QGeo	Yes	Low/medium	Yes
DRAMA-style MARL	Learned	High	Partially
Global teacher	Yes/learned	Not deployable	No
GLOBE++ Lite-P+	Yes/learned	Low	Yes

E. Knowledge Distillation and Policy Distillation

Knowledge distillation transfers behavior from a high-capacity teacher to a compact student. In this work, distillation is performed over next-hop action distributions rather than only over a single hard action. The soft teacher distribution carries relative preferences among candidate neighbors, which is valuable when several next hops are feasible but differ in stability, onward connectivity, and deadline risk.

F. Positioning of This Work

Table I summarizes the intended positioning. The strongest defensible claim is not that GLOBE++ Lite-P+ is the first to use global state, prediction, or RL. The defensible claim is that it uses global routing knowledge only during training, compresses it into a local execution policy, and then applies an interpretable predictive risk switch without online global commands.

III. Problem Formulation

A. Network Model

At time step t, the FANET is represented as a directed graph

$$\mathcal{G}_t = (\mathcal{V}, \mathcal{E}_t), \tag{1}$$

where $\mathcal{V} = \{1, \ldots, N\}$ is the set of UAVs and $(u, v) \in \mathcal{E}_t$ if UAV v is within the communication range R_c of UAV u and the link is available at time t. UAV u has position $p_u(t) \in \mathbb{R}^3$, velocity $v_u(t)$, queue state $q_u(t)$, and packet-level destination d.

For a packet currently at UAV u, the one-hop candidate set is

$$\mathcal{A}_u(t) = \mathcal{N}_u(t) \cup \{a_{\mathrm{DROP}}\}, \tag{2}$$

where $\mathcal{N}_u(t) = \{v : (u, v) \in \mathcal{E}_t\}$ and a_{DROP} is the explicit no-forward/drop action used by the simulator.

B. Partial Observation

The deployed router does not observe $\mathcal{G}_t$. It observes a local state

$$o_u(t) = \{x_u(t), x_i(t) : i \in \mathcal{N}_u(t), x_{u,i}(t), x_d(t)\}, \tag{3}$$

where x_u is the self feature vector, x_i is a neighbor feature vector, $x_{u,i}$ is an edge/candidate feature vector, and x_d encodes destination-related geometry. The global teacher may observe $\mathcal{G}_t$ during offline training, but the deployed policy must select

$$a_t \sim \pi_S(a \mid o_u(t)) \tag{4}$$

using only local observation.

C. Objective

The routing objective is to maximize reliable and timely delivery while controlling cost:

$$\max_{\pi} \mathbb{E}_\pi \left[R_{\text{succ}} - \alpha D - \beta E - \gamma C - \xi H\right], \tag{5}$$

where R_{succ} rewards packet delivery, D is delay, E is energy proxy, C is execution input/control cost, and H is path or hop penalty.

We report the following metrics:

$$\text{PDR} = \frac{N_{\text{delivered}}}{N_{\text{generated}}}, \tag{6}$$

$$\text{Deadline} = \frac{N_{\text{delivered before deadline}}}{N_{\text{generated}}}, \tag{7}$$

$$\bar{D} = \frac{1}{N_{\text{delivered}}} \sum_{k \in \mathcal{D}} (t_k^{\text{arr}} - t_k^{\text{src}}), \tag{8}$$

$$\text{Delay}_{95} = Q_{0.95}\left(\{t_k^{\text{arr}} - t_k^{\text{src}} : k \in \mathcal{D}\}\right), \tag{9}$$

$$\text{Throughput} = \frac{\sum_{k\in\mathcal{D}} B_k}{T_{\text{sim}}}, \tag{10}$$

$$\text{Overhead} = \frac{B_{\text{input/control}}}{N_{\text{delivered}} + \epsilon}, \tag{11}$$

$$\text{Energy} = \frac{1}{N_{\text{tx}}} \sum_{(u,v)} \left(\frac{\|p_u - p_v\|_2}{R_c}\right)^2. \tag{12}$$

Here $\mathcal{D}$ is the set of delivered packets, B_k is packet size, and ϵ prevents division by zero. Teacher-student transfer is measured by

$$D_{\text{KL}}(\pi_T \| \pi_S) = \sum_{a \in \mathcal{A}_u} \pi_T(a \mid s_t) \log \frac{\pi_T(a \mid s_t)}{\pi_S(a \mid o_u(t))}, \tag{13}$$

and by the return gap

$$\Delta J = J(\pi_T) - J(\pi_S). \tag{14}$$

IV. Proposed Method

A. Overview

Fig. 1 summarizes the proposed workflow. A privileged teacher is trained offline with global graph information. A student receives local observations and learns to imitate teacher action distributions. The deployed policy combines a nominal distilled branch with a predictive branch through a risk switch. The Phase 13 GLOBE++ Lite-P+ extension adds link-loss-aware risk, top-k onward stability, energy-aware tie breaking, and drop suppression.

B. Global Teacher

The teacher policy $\pi_T(a \mid s_t)$ observes global topology state s_t, including graph structure, node positions, mobility, queue features, and destination information. In the current implementation, the teacher is implemented as a global actor-critic model and trained with PPO during the earlier phases of the project. The teacher is not deployed. Its role is to generate routing behavior and soft targets:

$$y_T(a) = \text{softmax}\left(\frac{z_T(a)}{T}\right), \tag{15}$$

where $z_T(a)$ is the teacher logit for action a and T is the distillation temperature.

C. Local Student Distillation

The student policy $\pi_S(a \mid o_u)$ observes only local candidate features. The distillation objective combines soft teacher imitation, hard action imitation, and optional oracle/path targets:

$$\mathcal{L}_{\text{KD}} = \lambda_{\text{KL}} T^2 D_{\text{KL}}(y_T \| y_S) + \lambda_{\text{CE}} \text{CE}(a_T, \pi_S) + \lambda_{\text{O}} \mathcal{L}_{\text{oracle}}, \tag{16}$$

where $y_S = \text{softmax}(z_S/T)$, $a_T = \arg\max_a \pi_T(a \mid s_t)$, and $\mathcal{L}_{\text{oracle}}$ biases the student toward feasible shortest-path or recovery actions when such labels are available.

D. Predictive Candidate Features

For each candidate next hop i, the risk branch uses

$$x_i^{\text{risk}} = [m_i, \ell_i, q_i, o_i], \tag{17}$$

where m_i is link margin, ℓ_i is predicted current-link lifetime, q_i is queue headroom, and o_i is best onward-link lifetime after selecting candidate i.

The Phase 13 P+ extension adds

$$x_i^{+} = [\bar{o}_{i,\text{top}k}, \rho_i, p_{i,\text{keep}}, \eta_i], \tag{18}$$

where $\bar{o}_{i,\text{top}k}$ is the average of the top-k onward link lifetimes, ρ_i is normalized onward redundancy, $p_{i,\text{keep}}$ is link survival probability under stochastic link loss, and

$$\eta_i = 1 - \left(\frac{d_i}{R_c}\right)^2 \tag{19}$$

is an energy-efficiency proxy for distance d_i between the current UAV and candidate i.

E. Risk Score and Safety Gain

The P+ danger score is

$$\begin{aligned} D_i = {} & [g_m - m_i]_+ + [g_\ell - \ell_i]_+ + [g_o - o_i]_+ \\ & + [g_k - \bar{o}_{i,\text{top}k}]_+ + 0.5[g_\rho - \rho_i]_+ + [g_p - p_{i,\text{keep}}]_+, \end{aligned} \tag{20}$$

where $[x]_+ = \max(x, 0)$ and the g terms are safety gates.

IV-B. Reinforcement-Learning-Based Global Teacher Training

Reinforcement learning is used before deployment to train the privileged global teacher.
The deployed UAVs do not run PPO; they only execute the distilled local student and risk switch.

1) Global Markov state and teacher policy

$s_t = (G_t, p_t, v_t, q_t, d)$
$a_t \sim \pi_T(a_t \mid s_t; \theta_T)$
G_t: global FANET graph
p_t, v_t, q_t: position, velocity, queue state
d: packet destination

2) Routing reward

$r_t = \alpha_s R_{succ} - \alpha_D D_t - \alpha_E E_t$
$\quad - \alpha_H H_t - \alpha_F F_t$
R_{succ}: delivery reward
D_t, E_t, H_t: delay, energy, hop penalties
F_t: drop, loop, or route-failure penalty

The teacher maximizes expected discounted routing return:

$$J(\theta_T) = E_{\pi_T} \left[\sum_{t=0}^{T_{ep}} \gamma^t r_t \right]$$

PPO updates the teacher by comparing the new and old policies:

$$\rho_t(\theta_T) = \pi_T(a_t \mid s_t; \theta_T) \,/\, \pi_T(a_t \mid s_t; \theta_{T,old})$$
$$A_t = \hat{R}_t - V_T(s_t; \phi_T)$$

The clipped PPO objective limits unstable policy updates:

$$L_{PPO} = -E_t \left[\min\left(\rho_t(\theta_T) A_t,\; \mathrm{clip}(\rho_t(\theta_T), 1-\epsilon, 1+\epsilon) A_t \right) \right]$$

The value head and entropy regularizer are trained jointly:

$$L_V = E_t \left[(V_T(s_t; \phi_T) - \hat{R}_t)^2 \right]$$
$$L_T = L_{PPO} + c_V L_V - c_H H(\pi_T)$$

3) What the global teacher learns

The teacher is rewarded for long-term routing outcome,
not only for one-hop geographic progress.
It can learn to prefer:
- a slightly longer but more stable link
- a neighbor with onward connectivity
- a route avoiding drops, loops, and dead ends

4) Why the teacher is not deployed

The teacher uses s_t, which includes global topology.
That information is expensive or unavailable online.
Therefore only $\pi_S(\cdot \mid o_u; \theta_S^*)$ is deployed.
Each UAV executes local inference plus risk switching.
No PPO update is performed during packet forwarding.

After teacher convergence, the teacher is frozen and becomes a distillation target:

PPO teacher θ_T^* → **Teacher logits** $\pi_T(\cdot \mid s_t; \theta_T^*)$ → **KD loss** $D_{KL}(\pi_T \,\|\, \pi_S)$ → **Deploy student** $\pi_S(\cdot \mid o_u; \theta_S^*)$

Key deployment implication:

PPO and the global teacher are offline-only. During UAV operation, each node uses local observation o_u, the distilled student weights θ_S^*, and the risk-switch rule to select the next hop.

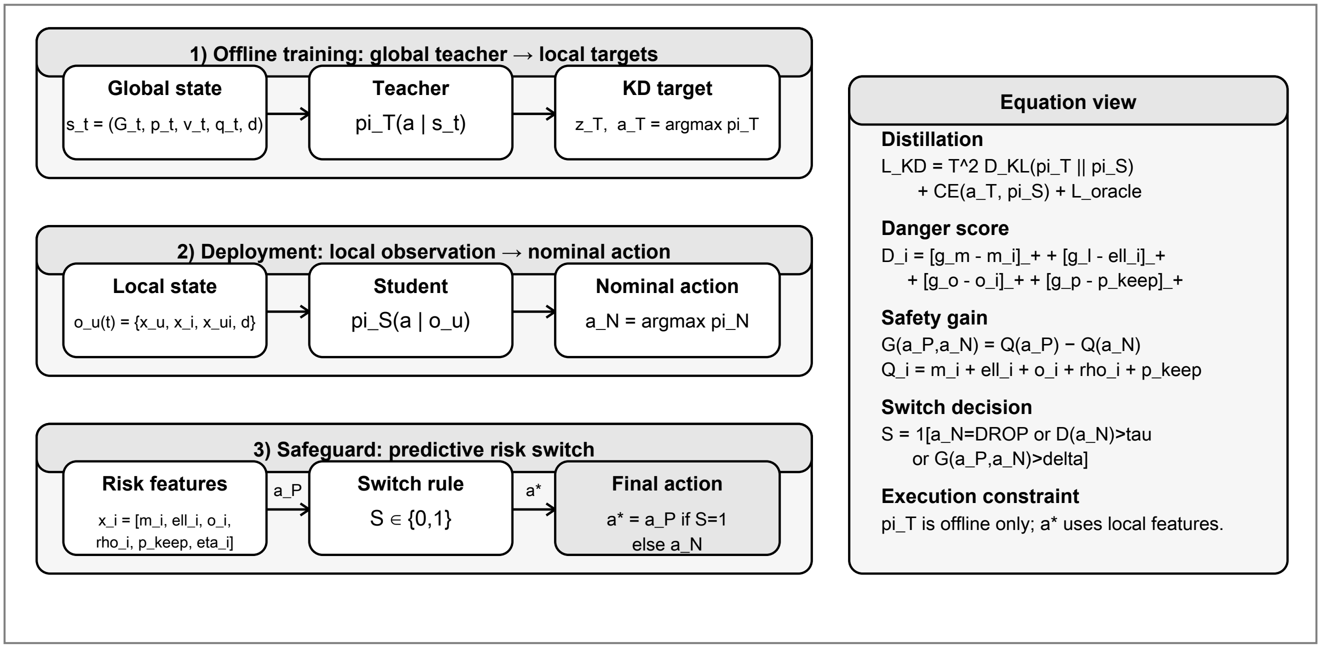

Fig. 1. System architecture of the proposed GLOBE++ Lite-P+ routing pipeline. The global teacher is used only for offline privileged training and distillation, while the deployed router selects the next hop from local one-hop observations through a nominal branch, a predictive branch, and a risk-switch decision.

The candidate safety utility is

$$Q_i = m_i + \ell_i + o_i + \bar{o}_{i,\mathrm{top}k} + 0.5\rho_i + p_{i,\mathrm{keep}}. \tag{21}$$

For a predictive action a_P and nominal action a_N, the safety gain is

$$G(a_P, a_N) = Q_{a_P} - Q_{a_N}. \tag{22}$$

F. Risk-Switch Decision

The nominal branch is used by default. The predictive branch is activated only when the nominal action is unsafe:

$$S = \mathbb{1}\left[a_N = a_{\mathrm{DROP}} \vee D_{a_N} > \tau \vee (a_P \neq a_N \wedge G(a_P, a_N) > \delta \wedge D_{a_P} < D_{a_N})\right]. \tag{23}$$

The final action is

$$a^\star = \begin{cases} a_P, & S = 1, \\ a_N, & S = 0. \end{cases} \tag{24}$$

G. Energy Tie Breaking and Drop Suppression

The predictive logit is adjusted by an energy tie term:

$$z_i^{P+} = z_i^P + \lambda_E \eta_i. \tag{25}$$

This term is intended to choose shorter and more energy-efficient links when safety is similar.

If at least one safe forwarding candidate exists, the drop logit is reduced:

$$z_{a_{\mathrm{DROP}}}^{P+} = z_{a_{\mathrm{DROP}}} - \lambda_D. \tag{26}$$

This does not force forwarding in all cases; it only discourages premature dropping when a locally safe candidate is available.

Algorithm 1 GLOBE++ Lite-P+ Local Routing Decision

Require: Local observation $o_u(t)$, candidate set $\mathcal{A}_u(t)$, nominal branch π_N, predictive branch π_P
Ensure: Selected next-hop action $a^\star$

```
for each candidate i ∈ N_u(t) do
    Compute m_i, ℓ_i, q_i, o_i
    Compute ō_{i,topk}, ρ_i, p_{i,keep}, η_i
    D_i ← [g_m − m_i]_+ + [g_ℓ − ℓ_i]_+ + [g_o − o_i]_+
          +[g_k − ō_{i,topk}]_+ + 0.5[g_ρ − ρ_i]_+ + [g_p − p_{i,keep}]_+
    Q_i ← m_i + ℓ_i + o_i + ō_{i,topk} + 0.5ρ_i + p_{i,keep}
    z_i^{P+} ← z_i^P + λ_E η_i
end for
if there exists a safe forwarding candidate then
    z_{a_DROP}^{P+} ← z_{a_DROP} − λ_D
end if
a_N ← arg max_a π_N(a | o_u(t))
a_P ← arg max_a π_P^+(a | o_u(t))
G ← Q_{a_P} − Q_{a_N}
if a_N = a_DROP or D_{a_N} > τ or (a_P ≠ a_N and G > δ
  and D_{a_P} < D_{a_N}) then
    a* ← a_P
else
    a* ← a_N
end if
return a*
```

V. Theoretical Analysis

A. Local Deployability

The deployed policy is locally executable if $a^\star$ can be computed from $o_u(t)$ and candidate-local predictions. In

GLOBE++ Lite-P+, m_i, ℓ_i, q_i, o_i, $\bar{o}_{i,\text{top}k}$, ρ_i, $p_{i,\text{keep}}$, and η_i are candidate-local or one-hop-derived quantities in the simulator. Therefore, the final action does not require $\mathcal{G}_t$ or online teacher inference.

B. Why Risk Switching Helps

Let a_N be the nominal local action and a_P be the predictive action. If D_{a_N} is low, the switch remains inactive and the policy behaves like the distilled nominal router. If D_{a_N} is high, the policy tests whether the predictive candidate has lower danger and sufficient safety gain. This makes switching selective:

$$\Pr(S = 1 \mid D_{a_N} \leq \tau) \approx 0, \tag{27}$$

while allowing recovery when the nominal action points to an unstable or dead-end candidate.

C. Distillation Error and Return Gap

The student can lose performance relative to the teacher due to partial observation and model compression. A useful diagnostic is the return gap

$$\Delta J = J(\pi_T) - J(\pi_S). \tag{28}$$

Reducing $D_{\text{KL}}(\pi_T \| \pi_S)$ does not guarantee a small return gap under distribution shift, because routing failures often occur in rare states such as holes or imminent link breaks. This motivates explicit scenario stress tests and risk-aware switching rather than relying only on average distillation loss.

D. Communication-Cost Claim

The main communication-cost claim is qualitative unless per-packet control signaling is measured. This draft currently reports local policy input bytes from the simulator, which is a proxy for execution-time information footprint. A submission-quality version should additionally report control bytes per delivered packet and compare against DRAMA-style message-passing baselines.

VI. Experimental Setup

A. Simulator and Scenarios

The Phase 12 full evaluation uses the local Lite-GLOBE simulator in ResearchAIWorkspace/implementations/lite_globe. The full run contains five training seeds, 14 evaluation scenarios, 200 evaluation episodes per scenario, six Phase 12 methods, 84,000 episode rows, 420 seed summary rows, and 1,764 statistics rows.

The scenario groups include general unseen mobility settings, routing-hole conditions, predictive-break conditions, and node-scalability sweeps. Predictive-break scenarios are especially important because they test whether a policy avoids links that look geographically attractive but are likely to break.

B. Compared Methods

The implemented comparison set is:

- GPSR: local greedy geographic routing baseline.
- Predictive Geographic: strong heuristic baseline using link prediction.
- Evo-QGeo: external RL-inspired predictive geographic baseline.
- IQMR Q(λ): external Q-learning-style routing baseline.
- DRAMA: external MARL/emergent-communication-style baseline.
- Phase 8 Geo-Residual KD: earlier distilled local student with geographic residuals.
- Lite-GLOBE-P variants: predictive-prior-only, no-switch, and Risk-Switch Lite-GLOBE-P.

AODV and OLSR-style classical ad hoc baselines are not implemented in the current codebase and should not be claimed as evaluated.

C. Evaluation Metrics

The primary reliability metrics are PDR and deadline delivery. Delay is reported with the 95th percentile to expose tail behavior. Efficiency is measured using an energy proxy and policy input bytes. Teacher-student training phases also track KL divergence and action agreement. Phase 13 P+ additionally logs switch steps and switch-related diagnostics.

D. Reproducibility Artifacts

The Phase 12 results used in this draft are stored under:

- ResearchAIWorkspace/artifacts/lite_globe/phase12/raw/
- ResearchAIWorkspace/artifacts/lite_globe/phase12/summaries/
- ResearchAIWorkspace/artifacts/lite_globe/phase12/tables/
- ResearchAIWorkspace/artifacts/lite_globe/phase12/figures/

The Phase 13 P+ implementation and smoke outputs exist, but full Phase 13 Colab validation is still missing in this draft.

VII. Results

A. Overall Performance

Table II reports the merged Phase 12 full results and external baseline comparison. The fully evaluated Risk-Switch Lite-GLOBE-P model achieves the highest overall PDR and deadline delivery. It also has lower delay p95 than Predictive Geographic, Evo-QGeo, IQMR Q(λ), and DRAMA, while using fewer input bytes than the predictive and message-passing external baselines.

B. Reliability

Risk-Switch Lite-GLOBE-P improves PDR by 32.5% over GPSR, 1.5% over Predictive Geographic, 2.1% over Evo-QGeo, and 1.6% over DRAMA. Its deadline delivery improvement follows the same trend, indicating that the method does not simply deliver more packets after excessive delay; it also improves timely delivery.

TABLE II
Overall average performance over the merged 14-scenario evaluation. Higher is better for PDR and Deadline. Lower is better for Delay p95, Energy, and Input bytes.

Method	PDR	Deadline	Delay p95	Energy	Input bytes
GPSR	0.683	0.637	4.163	1.166	2,284
Predictive Geographic	0.892	0.823	4.500	1.809	5,531
Evo-QGeo	0.887	0.815	4.557	1.812	6,452
IQMR Q(λ)	0.516	0.385	6.289	1.946	7,953
DRAMA	0.891	0.822	4.504	1.724	6,240
Phase 8 Geo-Residual KD	0.803	0.740	4.110	1.424	3,956
Lite-GLOBE-P predictive-prior only	0.905	0.838	4.334	1.776	5,424
Lite-GLOBE-P no-switch	0.890	0.822	4.300	1.726	5,775
Risk-Switch Lite-GLOBE-P	0.905	0.838	4.264	1.779	4,821

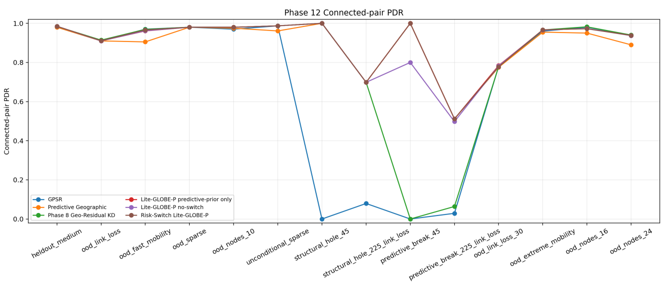

Fig. 2. Phase 12 PDR comparison across scenario groups and methods.

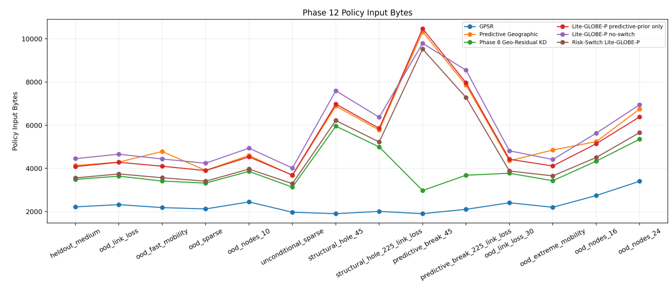

Fig. 4. Phase 12 local policy input-byte comparison.

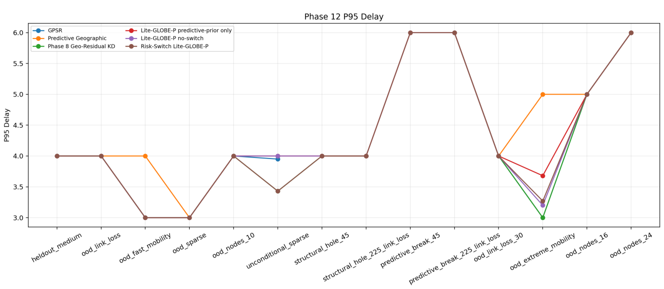

Fig. 3. Phase 12 95th-percentile delay comparison.

C. Delay

The overall delay p95 of Risk-Switch Lite-GLOBE-P is 4.264, lower than Predictive Geographic, Evo-QGeo, IQMR Q(λ), and DRAMA. Phase 8 has slightly lower delay p95, but Phase 8 fails in predictive-break scenarios. Therefore, delay must be interpreted jointly with PDR: a method that drops or fails early can appear fast while delivering fewer packets.

D. Input Footprint

Risk-Switch Lite-GLOBE-P uses more input bytes than GPSR and Phase 8, but fewer than Predictive Geographic, Evo-QGeo, and DRAMA. This supports the central deployment argument: the proposed family improves reliability without using the heavier execution-time footprint of predictive/message-passing external baselines. A stricter future version should report explicit control bytes per delivered packet.

E. Scenario-Level Findings

In routing-hole scenarios, Risk-Switch Lite-GLOBE-P matches the strong predictive and RL baselines while dramatically outperforming GPSR. In predictive-break scenarios, it repairs the major weakness of Phase 8 and matches Predictive Geographic, but Evo-QGeo remains stronger in one link-loss-heavy setting. This is the main motivation for the Phase 13 P+ extension.

F. Phase 13 Status

The current Phase 13 GLOBE++ Lite-P+ implementation is smoke-tested and includes switch diagnostics. However, the full Phase 13 results are not yet available. This draft therefore reports Phase 12 as the verified result set and treats Phase 13 P+ as the proposed final mechanism awaiting full validation.

VIII. Ablation Study

A. Implemented Ablations

The current project includes several internal ablations: KD-only student, KD+PPO student, Geo-Residual KD, Lite-GLOBE-P predictive-prior-only, Lite-GLOBE-P no-switch, and Risk-Switch Lite-GLOBE-P. These show that pure PPO fine-tuning did not provide consistent improvement, while geographic residuals and predictive risk features improved routing robustness.

B. Phase 13 P+ Planned Ablations

The Phase 13 implementation defines the following ablations:

- No link-loss gate: removes $p_{i,\mathrm{keep}}$ from the switch decision.
- No energy tie: removes $\lambda_E \eta_i$ from predictive logits.
- No drop suppression: removes the drop-logit penalty.
- Top-1 onward only: replaces top-k onward stability and redundancy with a single best onward estimate.

These ablations are necessary to prove that each P+ component contributes to the final behavior. At the moment, full multi-seed Phase 13 ablation evidence is still missing. The expected reviewer question is straightforward: if P+ improves performance, which component caused the improvement?

C. Metrics for Ablation

Each ablation should report PDR, deadline delivery, delay p95, energy proxy, input bytes, agent-drop rate, link-break failure rate, and switch activation rate. The switch rate is important because an always-on predictive branch would weaken the claim that the method is a selective local safeguard.

IX. Discussion

A. What the Current Evidence Supports

The Phase 12 evidence supports the claim that risk-switched global-to-local routing improves overall reliability and deadline delivery over the implemented baselines. It also supports the claim that the method can be more input-efficient than predictive and message-passing external baselines while retaining local execution.

B. What the Current Evidence Does Not Yet Prove

The current evidence does not yet prove that Phase 13 GLOBE++ Lite-P+ is superior to all baselines at full scale. It also does not prove superiority against unimplemented AODV/OLSR baselines, real-radio energy models, or physical testbed conditions. Claims about communication overhead should be strengthened with explicit control-byte measurements rather than only input-byte proxies.

C. Main Limitation

The most important technical limitation is the predictive-break-with-link-loss case where Evo-QGeo outperforms Risk-Switch Lite-GLOBE-P. This suggests that the current risk switch needs stronger link survival modeling and better treatment of second-hop route continuity. Phase 13 P+ addresses this directly through $p_{i,\mathrm{keep}}$, top-k onward stability, and redundancy ρ_i.

D. Submission Strategy

For a strong SCI-level submission, the paper should position GLOBE++ Lite-P+ as a deployable global-to-local routing distillation method rather than as a generic MARL or global-context method. The core story should be: global graph knowledge is useful, but only as an offline teacher; the deployed policy is local, selective, and measurable in overhead.

X. Conclusion

This paper draft presented GLOBE++ Lite-P+, a decentralized FANET routing framework that combines global-to-local policy distillation with predictive risk switching. The fully evaluated Phase 12 predecessor, Risk-Switch Lite-GLOBE-P, achieves the strongest overall PDR and deadline delivery among the implemented comparison set while reducing input footprint relative to predictive and message-passing external baselines. The Phase 13 P+ extension introduces link-loss-aware switching, top-k onward stability, energy-aware tie breaking, and drop suppression to address the remaining weaknesses found in Phase 12.

The next required step is full Phase 13 validation across the same 14-scenario, five-seed protocol, followed by component ablations and explicit communication-overhead reporting. Until then, Phase 12 results should be treated as verified evidence and Phase 13 P+ as the final proposed mechanism under validation.

References

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